KhLan (ខ្ឡាន) PROD AUDIT V0.2.1

របាយការណ៍សវនកម្មប្រព័ន្ធ និងការត្រួតពិនិត្យមុខងារពេញលេញ (System QA Audit & Feature Verification)

OFFICIAL QA AUDIT PASSED
Execution Date: November 5, 2026
Test Framework: Vitest 5.0 + Playwright 1.64

249 / 249	31	6 / 6	< 2.0s
TESTS PASSING	TEST SUITES	CORE PILLARS	SUITE LATENCY
100% Pass Rate (0 Failures)	Unit, Integration & E2E	100% Functional Coverage	High Velocity Test Pipeline

System Architecture & Service Topology

Dual-tier architecture engineered for high concurrency, payment security, and localized Cambodian banking:

Next.js 16 Edge
React 19, App Router, SSR, SEO Metadata, Tailwind CSS

→

Spring Boot / Node
Prisma ORM, SQLite / PostgreSQL, Session Guards

→

Bakong KHQR & OCR
NBC KHQR Strings, Tesseract.js Slip Engine

→

Storage & CDN
MinIO Primary + Cloudflare R2 Sync Failover

6-Pillar System Feature Verification Matrix

Pillar 1: Buyer Discovery, Search & Vehicle Comparison (ការស្វែងរក និងប្រៀបធៀបយានយន្ត)

38 Tests — ● PASSED

Multi-category catalog indexing (Cars, Motorcycles, Spare Parts), multi-parameter filtering, dependent make-model cascades, side-by-side comparison matrix, and financial amortization calculator.

- ✓ Multi-category schema filtering (Cars, Motorcycles, Spare Parts)
- ✓ Dependent Make -> Model cascade resolution (NHTSA vPIC & local)
- ✓ Side-by-side vehicle comparison with delta calculation
- ✓ Cambodia vehicle financing calculator (Down payment %, term, monthly USD)
- ✓ SEO server-rendered route crawling with JSON-LD schema (Car, Motor, Product)

Pillar 2: AI Vehicle Inspection & Valuation Engine (ប្រព័ន្ធ AI ត្រួតពិនិត្យឡាន និងវាយតម្លៃទីផ្សារ)

36 Tests

● PASSED

Evidence-based AI vehicle analysis evaluating chassis alignment, paint overspray, odometer rollback heuristics (4,000 - 25,000 km/yr), multi-document authenticity, and fair market price comps.

- ✓ AI Inspection confidence scoring engine (0–100% weighted confidence)
- ✓ Chassis alignment, body panel and paint overspray AI checks
- ✓ Market pricing comp engine with category-tailored transparency disclaimers
- ✓ Odometer consistency analysis tailored to Cambodian driving heuristics
- ✓ Multi-document validation (Registration book, Import declaration, Service records, Invoices)

Pillar 3: Strict 2–Party Verified Direct Chat & Anti-Scam Security (ប្រព័ន្ធជជែកផ្ទាល់ ២ ភាគី និងសុវត្ថិភាពការពារការបោកប្រាស់)

22 Tests

● PASSED

End-to-end verified communication between verified buyers and vehicle listing owners. Zero third-party data leakage, bot impersonation blocking, and in-chat bilingual scam warnings.

- ✓ Strict 2–party access control (403 forbidden for non-parties, 401 for guests)
- ✓ Anti-scam wire deposit warning banners in Khmer & English
- ✓ Vehicle listing context attachment with direct negotiation tools
- ✓ Zero third-party leakage guarantee across conversation threads
- ✓ Real-time auto-polling & message delivery synchronization

Pillar 4: Seller Subscription, Bakong KHQR & Bank Slip OCR Verification (ការជាវ គណនីអ្នកលក់ Bakong KHQR និង Tesseract OCR)

48 Tests | PASSED

Automated seller onboarding with 14-day trial management, dynamic NBC Bakong KHQR generation, and high-accuracy Tesseract OCR parsing across major Cambodian banks with Telegram admin fallback.

- ✓ Canonical seller plan tiers (FREE 3 cars, BASIC 15 cars, PRO 50 cars)
- ✓ NBC Bakong KHQR dynamic payload generation (USD & KHR currencies)
- ✓ Duplicate slip reuse prevention & Telegram admin review routing
- ✓ 14-day free trial lifecycle and quota enforcement guards
- ✓ Multi-bank OCR receipt parser (ABA Bank, ACLEDA, Wing Bank, Canadia)

Pillar 5: Multi-Tenant Dealership Management & Staff Roles (ការគ្រប់គ្រងយានដ្ឋានច្រើនសាខា និងសិទ្ធិបុគ្គលិក)

45 Tests | PASSED

Enterprise multi-tenant dealership management featuring role-based access control (OWNER, MANAGER, SALES), plan-based seat limits, multi-branch showrooms, and inventory stock tracking.

- ✓ Role-Based Access Control (RBAC): OWNER, MANAGER, SALES
- ✓ Showroom branch management and staff assignment
- ✓ Dealer pricing tier customization and bulk inventory batch operations
- ✓ Seat quota enforcement based on active subscription tier
- ✓ Inventory lifecycle management (DRAFT -> ACTIVE -> RESERVED -> SOLD)

Pillar 6: Admin Governance, Storage Resilience & Platform Safeguards (ការគ្រប់គ្រងរដ្ឋបាល ប្រព័ន្ធការពារទិន្នន័យ និងសុវត្ថិភាព)

60 Tests | PASSED

Centralized admin management console, inventory GMV analytics, primary MinIO to Cloudflare R2 storage failover synchronization, Nacos dynamic configuration, and Cloudflare Turnstile bot shielding.

- ✓ Platform metrics aggregation (Active listings, GMV volume, lead inquiries)
- ✓ Nacos dynamic configuration sync for distributed runtime configs
- ✓ Seller moderation, suspicious user suspension, and listing compliance audits
- ✓ Storage resilience engine: primary MinIO with Cloudflare R2 backup failover
- ✓ Cloudflare Turnstile token validation and brute-force protection

Automated Fraud Safeguards & Security Governance

KhLan enforces cryptographic, visual, and network-level protections against automotive and financial fraud in Cambodia:

🛡️ Strict 2-Party Isolation

Direct 1-on-1 messaging verified cryptographically. Third-party intercept attempts receive 403 Forbidden with zero chat leakage.

⚡ Slip Replay Attack Protection

Bank transfer receipts are indexed by transaction reference hash. Duplicate or altered slips are rejected automatically.

🔍 AI Odometer Anomaly Detection

Compares declared mileage against vehicle manufacturing year using 4,000-25,000 km/year

heuristics to flag rollback tampering.

KhLan Platform Engineering & QA Assurance
Phnom Penh, Kingdom of Cambodia · Built for Trust & Transparency

✓ PRODUCTION-READY VERIFIED